

Tigran Melkonyan

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RESEARCH INTERESTS

Hyperspectral image classification under limited data regimes (few-shot learning); control theory applications in multi-agent drone systems, including real-time synchronization and distributed coordination.

EDUCATION

Main

American University of Armenia [\[link\]](#), [\[transcript\]](#)
Bachelor of Science in Computer Science

Yerevan, Armenia
Aug. 2023 – Present

Yerevan State University [\[link\]](#), [\[transcript\]](#)
Informatics and Applied Mathematics

Yerevan, Armenia
Sep. 2022 – Jun. 2023

Armavir High School № 1 named after Ruben Yeghoyan [\[link\]](#)

Armavir, Armenia
Sep. 2019 – May. 2022

Araks Secondary School named after Zorik Avetisyan [\[link\]](#)

Armavir, Armenia
Sep. 2010 – May. 2019

Other courses

Mathematical Analysis III, Moscow State University (Online)
Supervisor - Stanislav Shaposhnikov

Sep. 2023 – Dec 2023

Analytic Geometry, Yerevan State University

Sep. 2022 – Dec. 2022

OLYMPIADS

The ICPC International Collegiate Programming Contest (Northern Eurasia Finals) Dec, 2024, [\[PDF\]](#), [\[PDF\]](#)

The ICPC International Collegiate Programming Contest (Armenia, Regional Contest), Nov, 2024 [\[PDF\]](#), [\[PDF\]](#)

Mathematics Olympiad held at Yerevan State University, 2022 [\[PDF\]](#)

PROFESSIONAL EXPERIENCE

Center for Scientific Innovation and Education [\[link\]](#)
Research Intern

Yerevan, Armenia
Mar. 2025 – Present

- Contributing to experimental research on communication delays, and control-theoretic methods for synchronized drone operations.

PUBLICATIONS AND MANUSCRIPTS

T. Melkonyan, A. Petrosyan, D. Hakobyan, S. Kumar.

A Comparative Analysis of Lung Nodule Detection using Transfer Learning, 2025.

Manuscript submitted.

—Conducted a full experimental study combining CNN-based classification and Faster R-CNN detection for lung nodule analysis on ChestX-ray14 and JSRT datasets.